SeeWandel Climate

Interreg
Alpenrhein-Bodensee-Hochrhein

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Regime Shifts in Plankton Dynamics: What Drives Them and Can We Predict Them?

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Ecomeets @ Eawag, Dübendorf
28 October 2026

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Slides, code and data shared openly
leomarameo7.github.io/science_communication.html

Acknowledgements



Pomati lab
Plankton Ecology, Eawag



Steve Munch & Tanya Rogers
EDM workshop guests

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Take home message

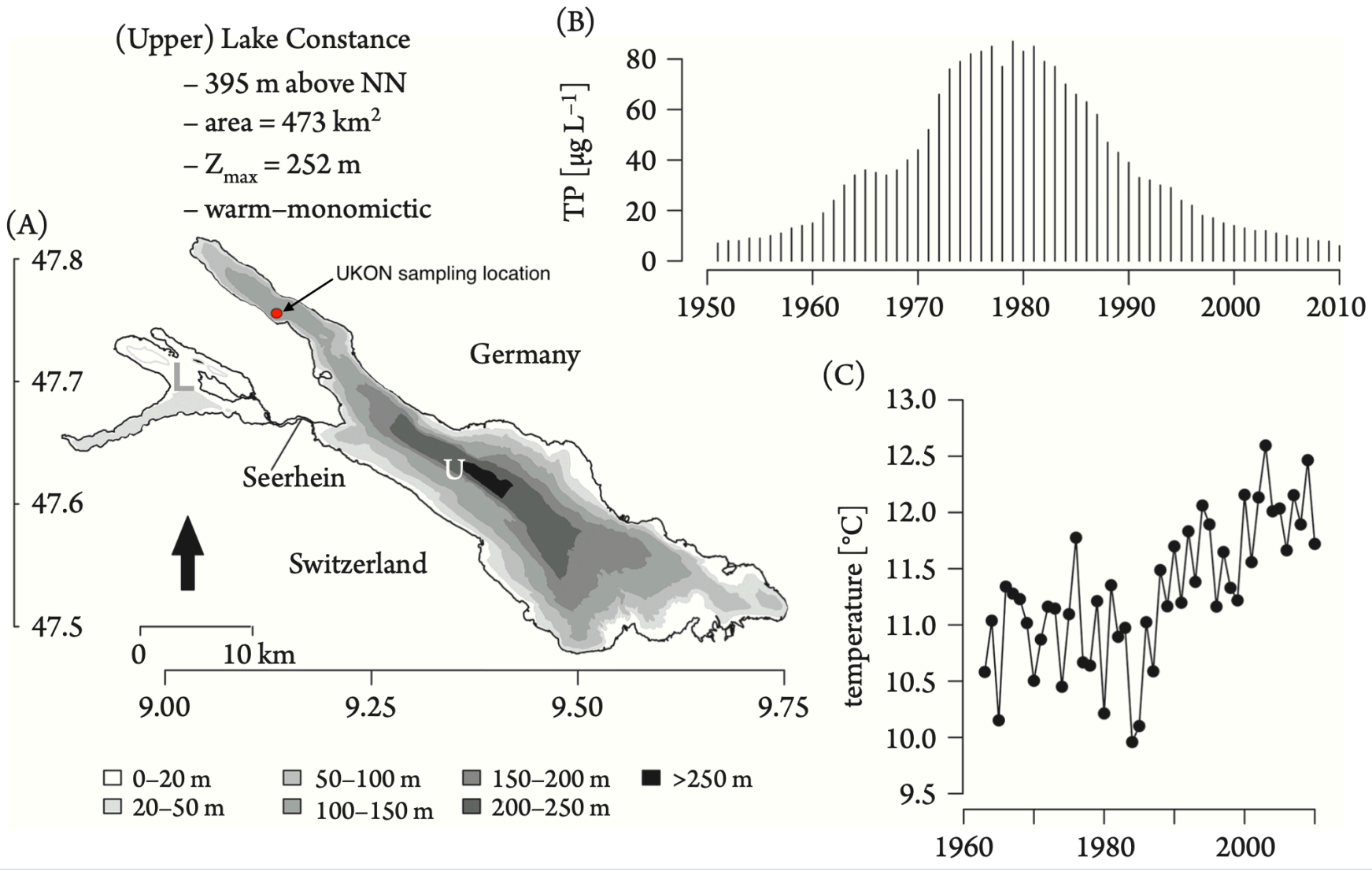
Plankton regime shifts can be detected and partly explained by biotic drivers. Whether they propagate and can be predicted is still open.



Outline

1. What we know: background
2. Research questions
3. Methods: our pipeline
4. Results: detection of regime shifts
5. Explain regime shifts
6. Propagation of regime shifts
7. Conclusions

What we know: background

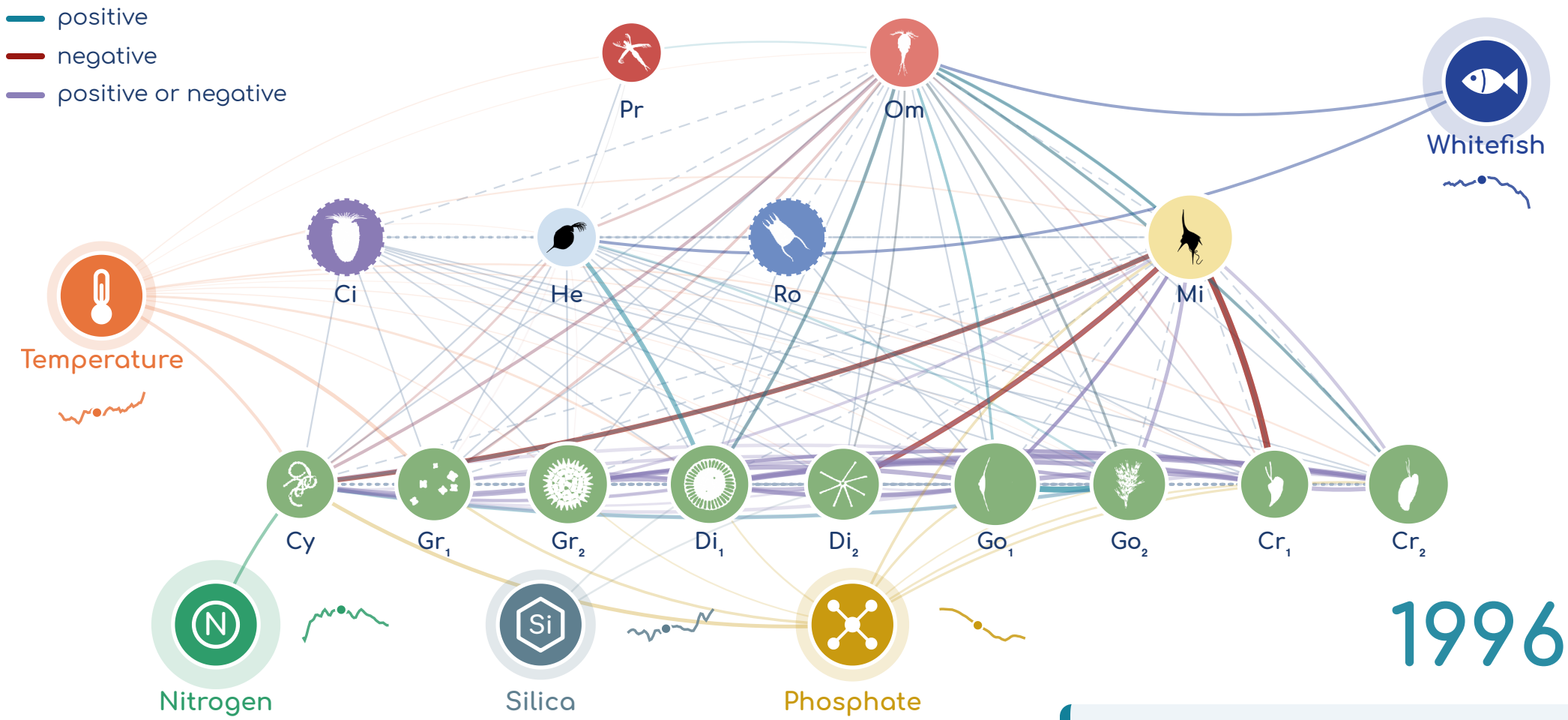


Lake Constance
473 km² · 252 m deep

Phosphate ↑ until 1980, then ↓

Temperature ↑ about 1 °C

What we know: the plankton community



Study period: **1975–2022** · monthly time series

The plankton community changes across time

Network modified from Merz et al. 2023, Nat. Clim. Change.

Research questions



1 · Detect

Abrupt shifts?



2 · Explain

What drives them?



3 · Propagate

Does a shift spread?



4 · Predict

Can we forecast them? (in progress)

Methods: our pipeline

1

Data

Monthly data 1975–2022

2

Detect

BEAST + two criteria

3

Explain

GP-EDM + hypotheses

4

Propagate

Shift-removal test

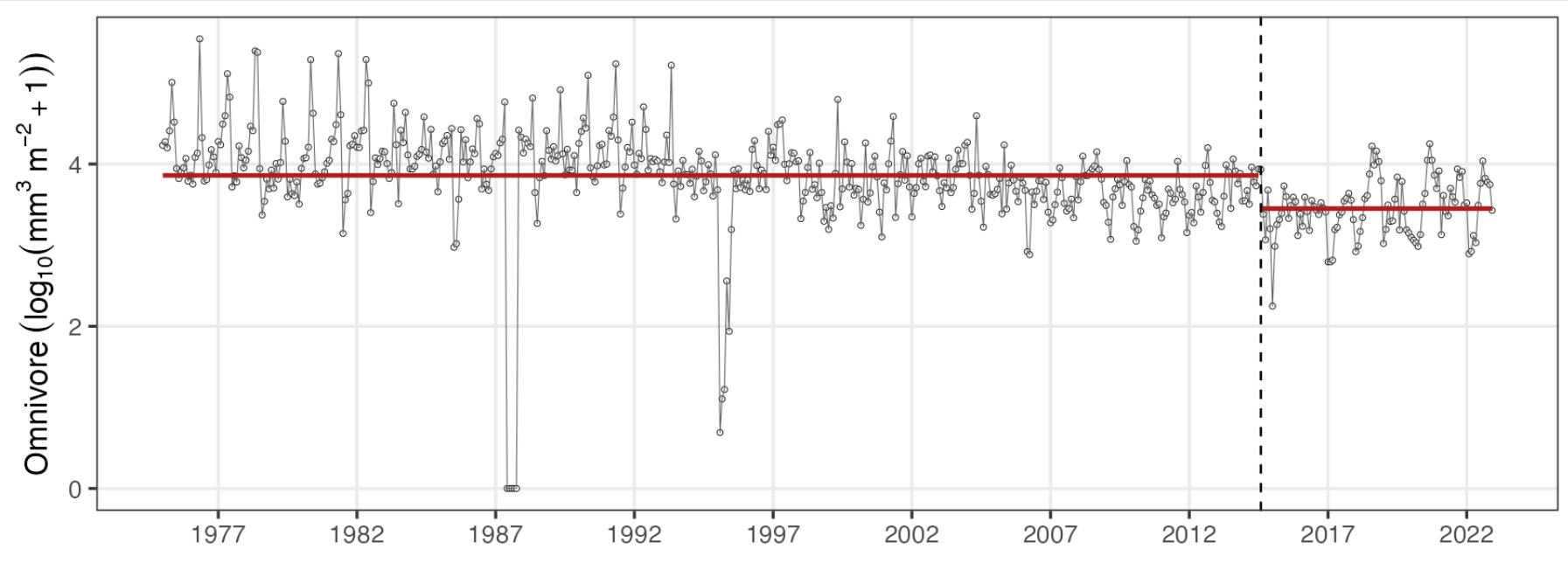
5

Predict

Forward validation

WORK IN PROGRESS

Results: detection of regime shifts



Omnivores, monthly biovolume. Dashed: shift; red: period means.

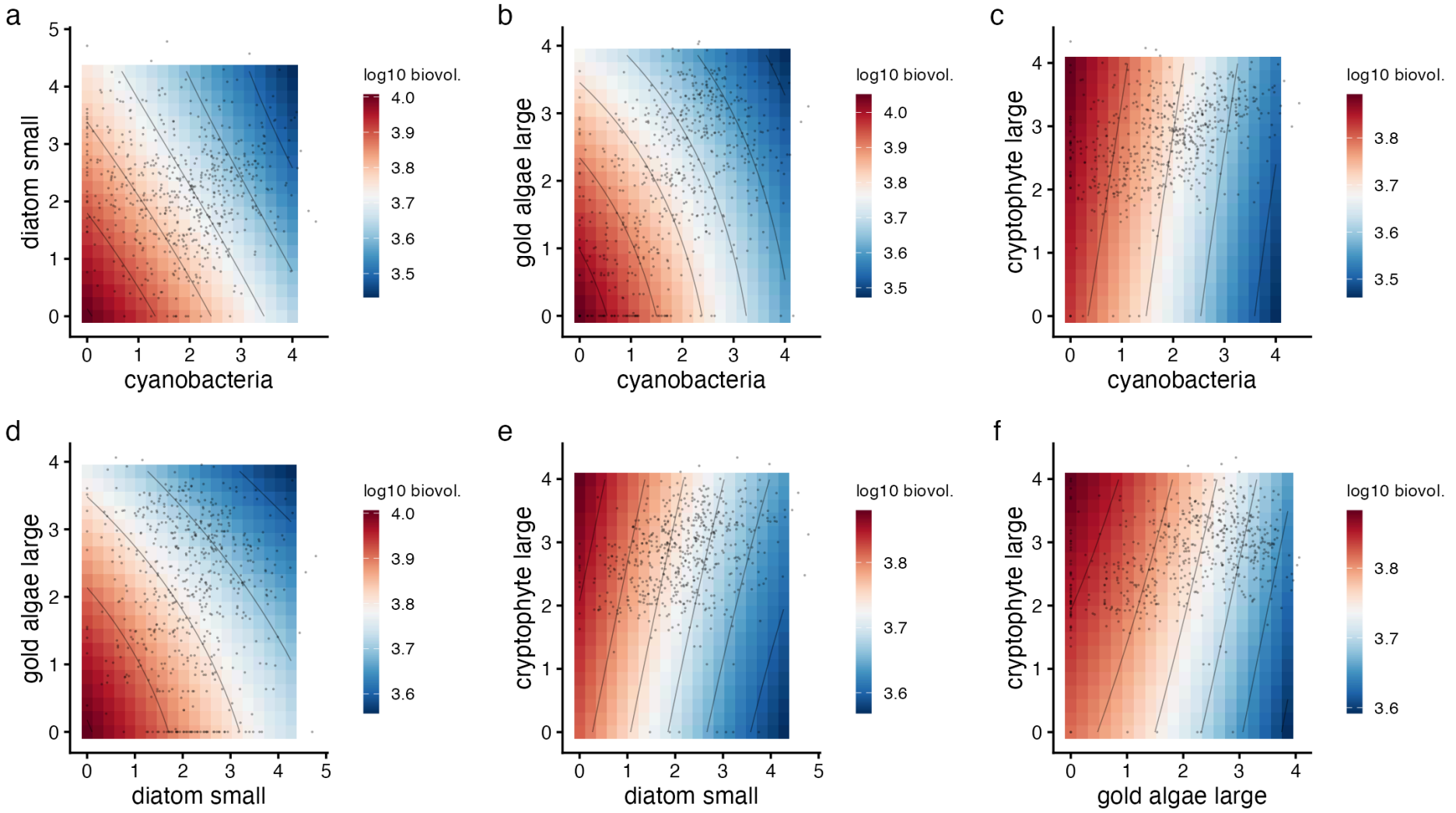


Shift: Aug 2014

abruptness 1.19

The omnivores dropped abruptly in 2014

Explain regime shifts



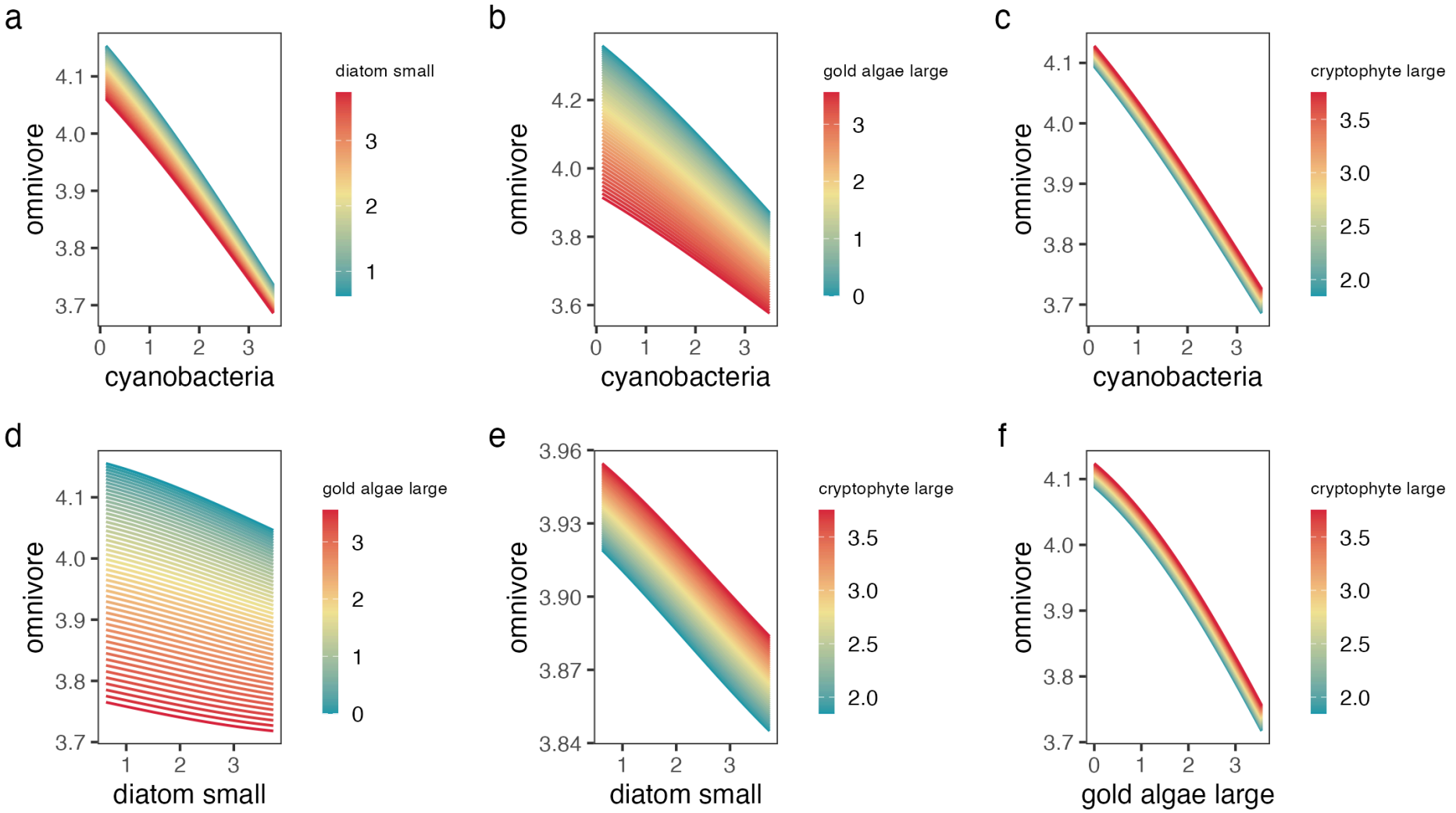


LOO R^2 = **0.81**

Omnivores ↓ with cyanobacteria, small diatoms, large gold algae

Prey hypothesis: **not supported**

Explain regime shifts: partial dependence





x: predictor 1
lines: predictor 2

Cyanobacteria ↓ omnivores

Lines never cross

Propagation of regime shifts

1 · Real predictor



2 candidate pairs

0 testable

Predictor shifted **after**

2 · Shift-free surrogate (× 100)



Gap between real and surrogate prediction = evidence for propagation

No evidence yet: too few shifts

Conclusions



Shifts are detectable



Biotic drivers matter



Propagation, prediction: open

Take home message

Plankton regime shifts can be detected and partly explained by biotic drivers. Whether they propagate and can be predicted is still open.



Acknowledgements

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igkb
Internationale Gewässerschutzkommission für den Bodensee

IBKF

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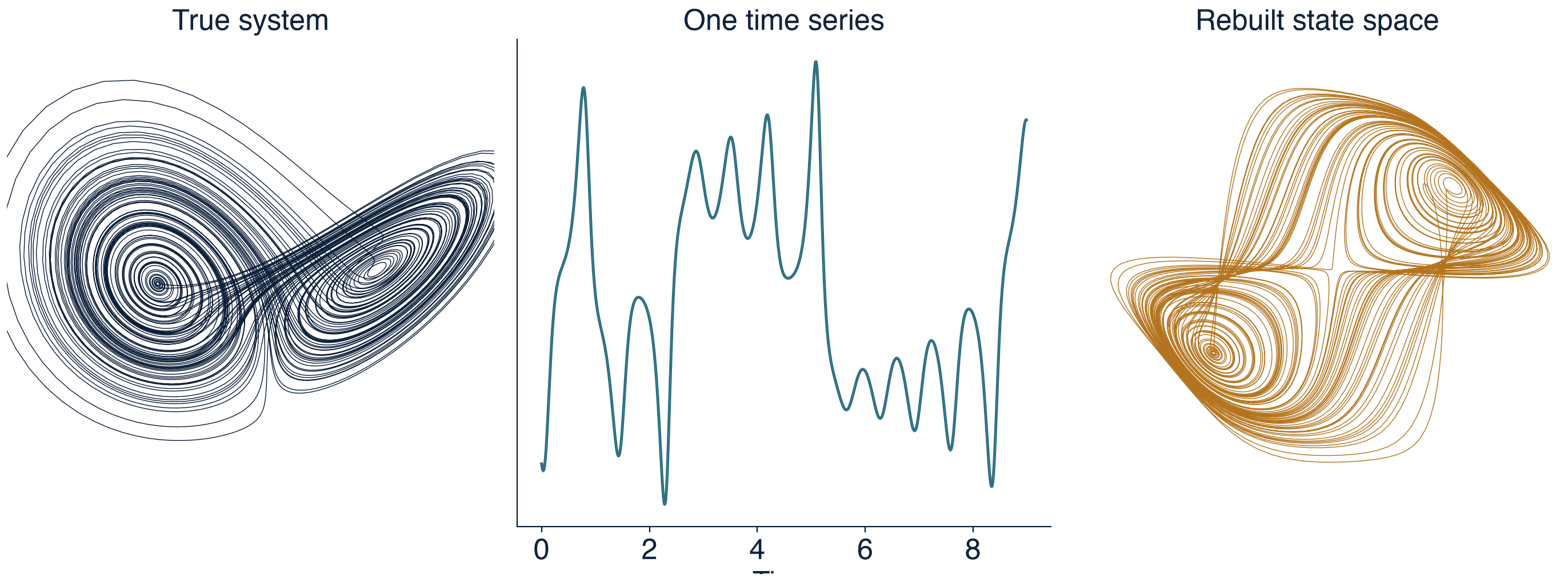
WSL

SeeWandel-Climate · Pomati lab and colleagues at Eawag
Luis J. Gilarranz · Stefanie Eyring · Stephan B. Munch · Tanya L. Rogers · Francesco Pomati
Data: IGKB, IBKF, University of Konstanz (D. Straile)

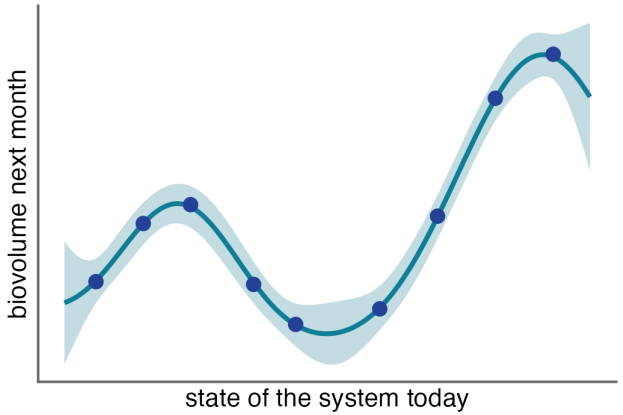
Supplementary materials



What is GP-EDM?



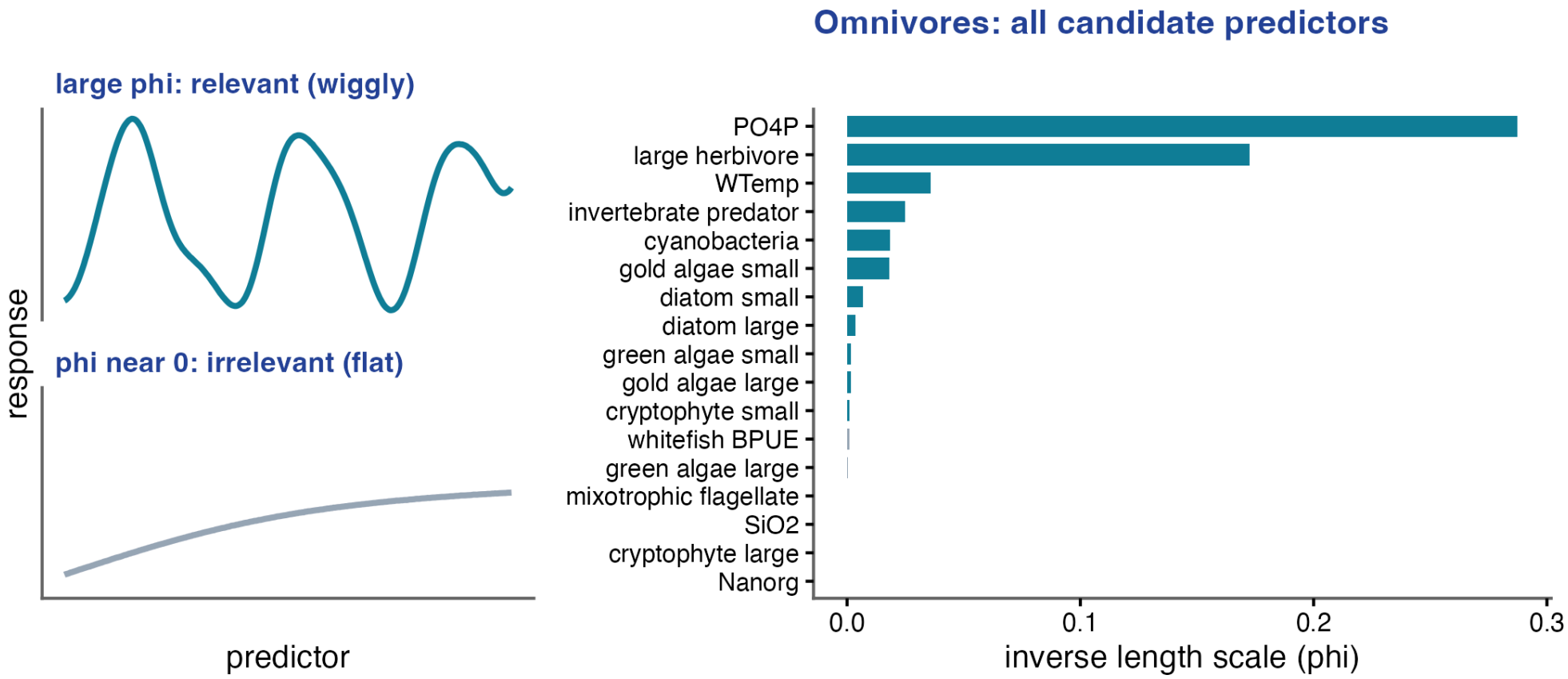
Gaussian process: no fixed equation



Next month = f(past states, drivers)

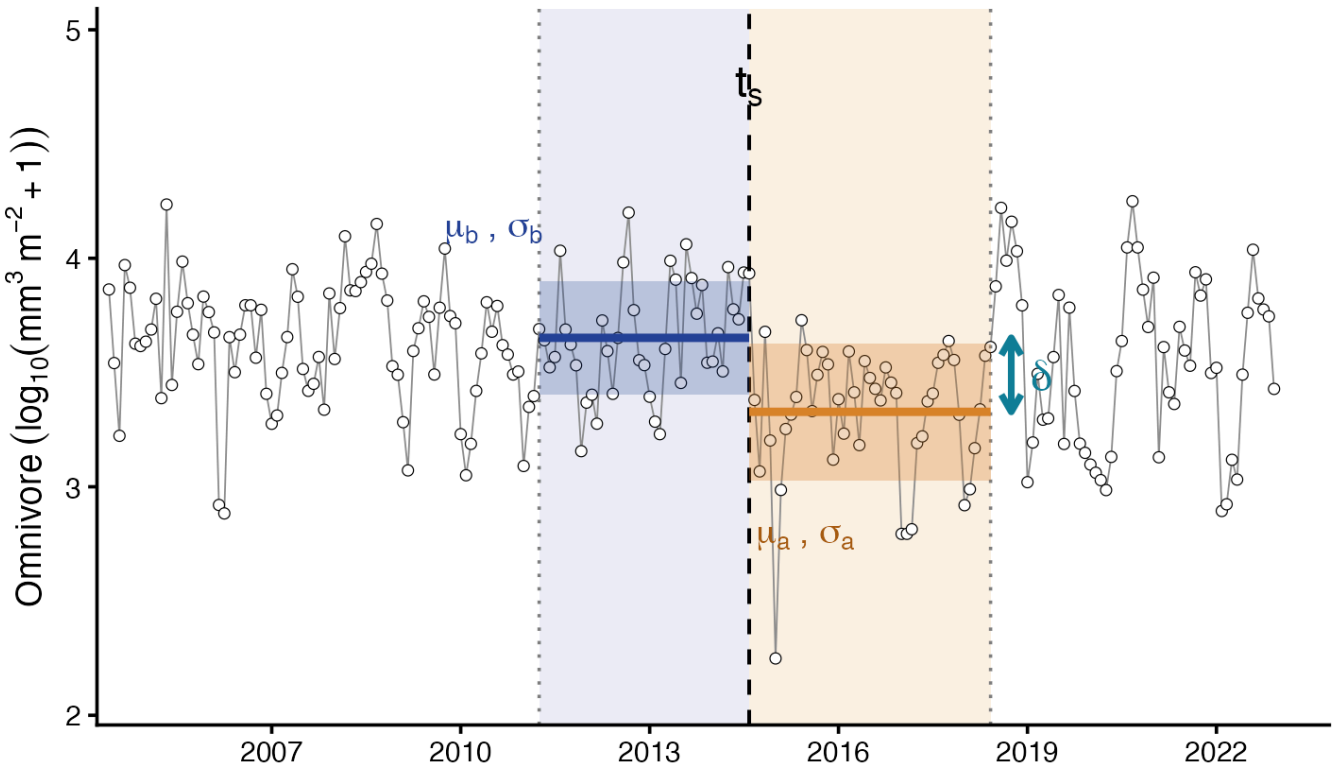
f learned from the data: Gaussian process

Automatic relevance determination (ARD)



One length scale per predictor: **phi near 0** = predictor dropped

Detecting regime shifts: BEAST and two criteria



1 BEAST: candidate shift

t_s : Pr(change point) = 0.87

2 Difference of the two chunks

$\delta = \mu_a - \mu_b \quad 0 \notin CI_{95\%}(\delta)$

CI = [0.19, 0.42] ✓

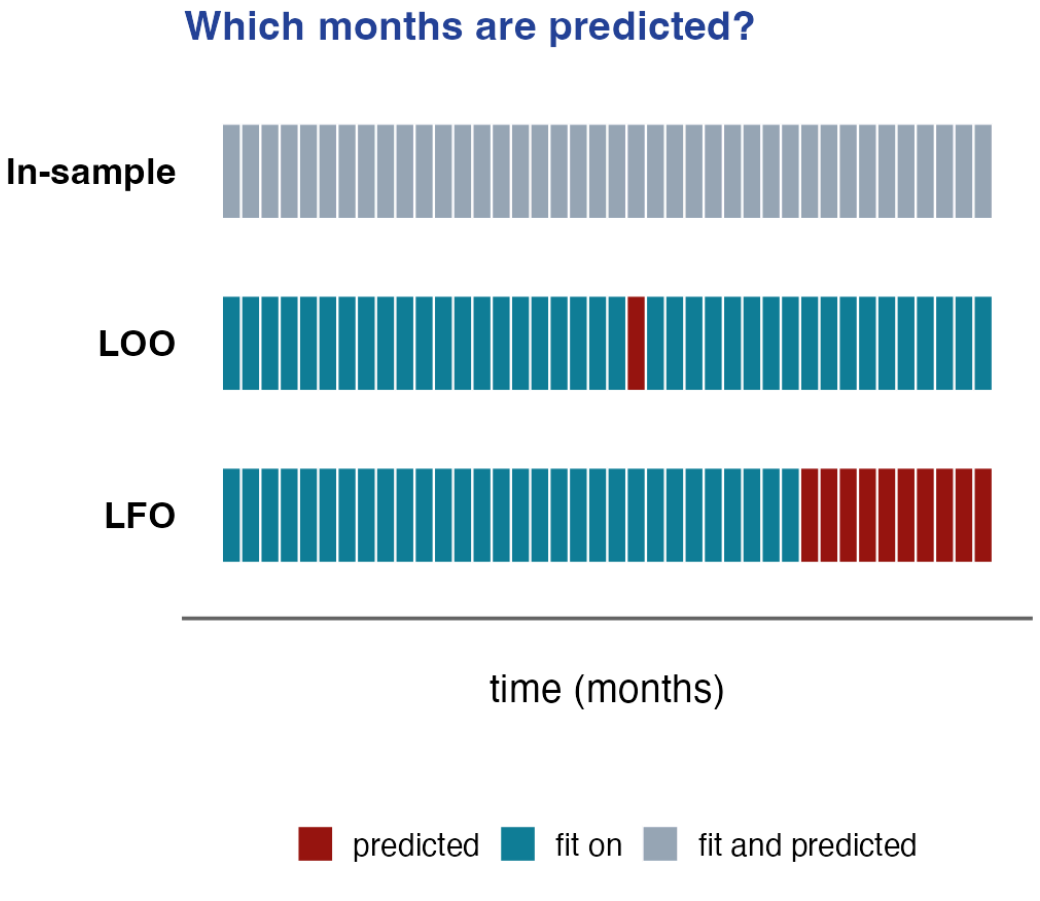
3 Abruptness

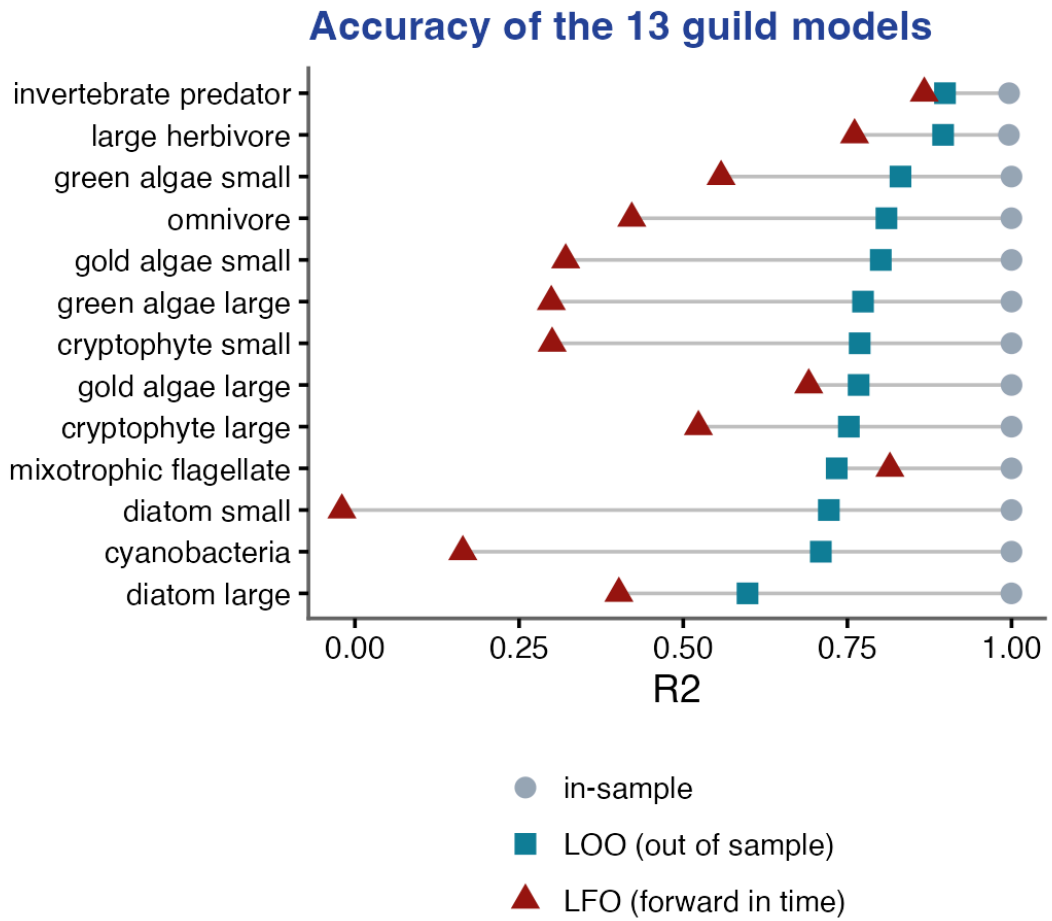
$$a = \frac{|\mu_a - \mu_b|}{\frac{1}{2}(\sigma_b + \sigma_a)} > 1$$

a = 1.19 ✓

Kept if 2 and 3 hold

In-sample versus out-of-sample R^2





In-sample: $R^2 = 1$, a GP interpolates

LOO: one month left out

LFO: forecast the last 10 %

Which genera? Zooplankton and mixotrophs



Invertebrate predators

Pr

Leptodora	73%
Bythotrephes	27%



Omnivores

Om

Cyclops	80%
Mesocyclops	17%
Acanthocyclops	1%
Thermocyclops	1%
Asplanchna	0%

+ 4 more



Large herbivores

He

Daphnia	91%
Eudiaptomus	6%
Bosmina	2%
Diaphanosoma	1%
Acanthodiaptomus	0%

+ 6 more



Mixotrophic flagellates

Mi

Ceratium	46%
Gymnodinium	45%
Peridinium	7%
other Flagellaten	1%
other Phytoflagellaten	0%

+ 14 more

Top genera by share of biovolume, IGKB 1975–2022

Which genera? Phytoplankton (1/2)



Cyanobacteria

Cy colonial

Anabaena	44%
Aphanizomenon	16%
Planktothrix	9%
Cyanothece	7%
Aphanothece	6%

+ 25 more



Green algae, small

Gr1 ≤ 550 µm³

Volvocales	30%
unidentified algae	24%
Chlorella	21%
Oocystis	7%
Characium	3%

+ 21 more

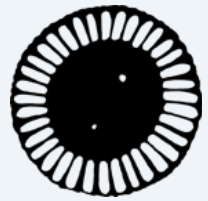


Green algae, large

Gr2 > 550 µm³ or colonial

other Chlorococcales	23%
Pandorina	18%
Mougeotia	16%
Chlamydomonas	10%
Ulothrix	6%

+ 46 more

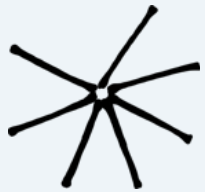


Diatoms, small

Di1 ≤ 800 µm³

Stephanodiscus	81%
Cyclotella	17%
Achnanthes	1%
Fragilaria	1%
Nitzschia	0%

+ 9 more



Diatoms, large

Di2 > 800 µm³ or colonial

Fragilaria	50%
Stephanodiscus	15%
Asterionella	14%
Cyclotella	10%
Tabellaria	4%

+ 18 more

Top genera by share of biovolume, IGKB 1975–2022

Which genera? Phytoplankton (2/2)



Gold algae, small

Go1 ≤ 250 µm³

Erkenia	63%
Chrysochromulina	23%
Chromulina	4%
Ochromonas	2%
Pseudokephyrion	2%

+ 13 more



Gold algae, large

Go2 > 250 µm³ or colonial

Dinobryon	42%
Mallomonas	17%
Uroglena	12%
Chrysophyceae	12%
Pseudopedinella	6%

+ 9 more



Cryptophytes, small

Cr1 ≤ 700 µm³

Rhodomonas	99%
Plagioselmis	1%
Cyathomonas	0%
Cryptaulax	0%
Chilomonas	0%

+ 1 more



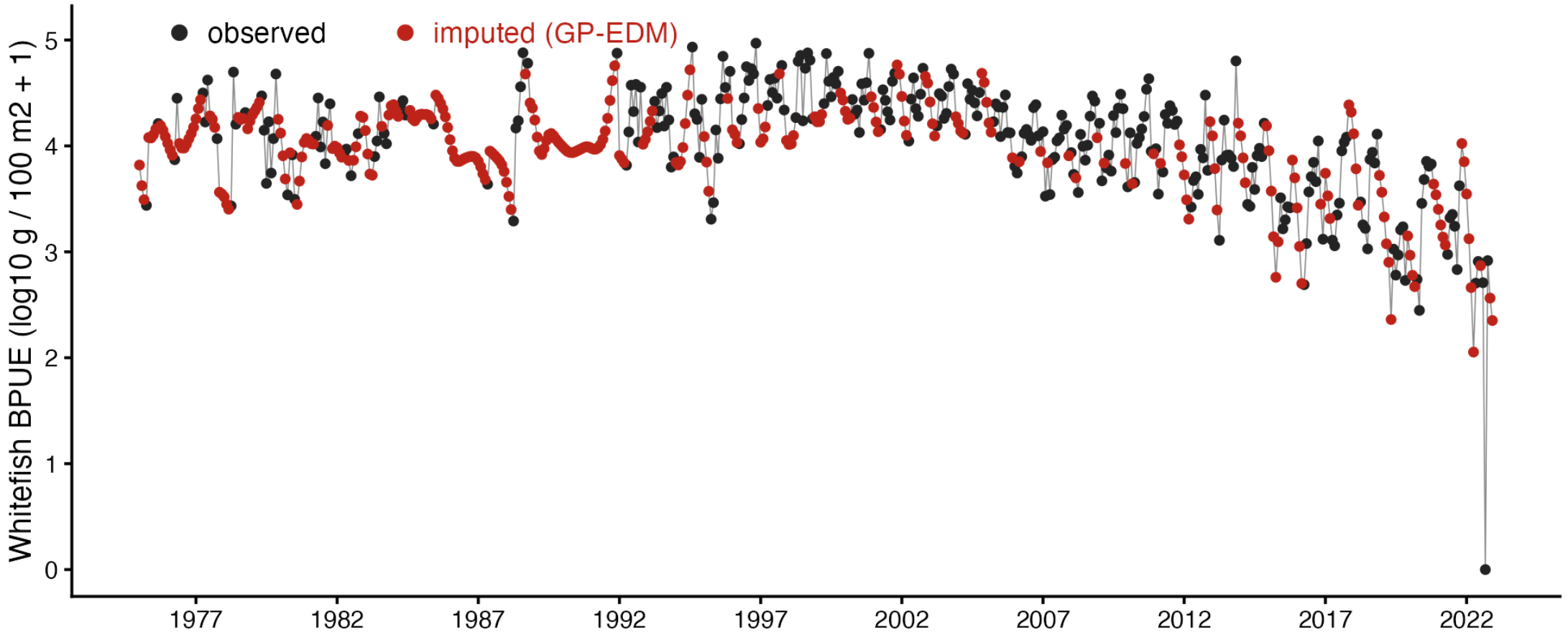
Cryptophytes, large

Cr2 > 700 µm³

Cryptomonas	100%
other Cryptophyceen	0%

Top genera by share of biovolume, IGKB 1975–2022

Whitefish time series: observed and imputed



287 of 576 months imputed (50 %)

GP-EDM, LOO R^2 = 0.55

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Figure of Lake Constance (map, phosphorus, temperature): D. Straile, University of Konstanz. Data: IGKB, IBKF.